

Hong Vinh Vu

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Education

Hanoi University of Science and Technology , Control and Automation Engineering	Hanoi, Vietnam
- Description 1. - Description 2.	2024 – 2028

Experience

Multi-Agent System Control (MASC) Research Laboratory, SEEE, HUST. , Research Assistant	HUST, VN
Applied Optimal Control Algorithms and Intelligent Control Algorithms to optimize trajectory planning for Quadrotors.	2026 – 2028
Developed a hybrid flight control framework by integrating PyTorch-based Deep Reinforcement Learning (DRL) for high-level dynamic trajectory planning with Model Predictive Control (MPC) and ANFIS for optimal trajectory tracking and nonlinear disturbance compensation.	2 years
Mechatronics Engineering Group (MEG), Motion Control and Applied Robotics Laboratory (MoCAR LAB), SEEE, HUST. , Research Assistant	HUST, VN
Kinematic Modeling and Control of a 6WD-6WS Mobile Robot based on ROS2.	2025 – 2026
Developed mathematical kinematic/dynamic models and motion control algorithms for a 6WD-6WS (6-Wheel Drive, 6-Wheel Steering) omnidirectional vehicle, implementing custom ROS2 navigation nodes, Gazebo physics simulations, and embedded MCU firmware for real-time trajectory tracking and multi-mode steering coordination.	1 year

Volunteer

People's Climate March , Lead Organizer	Zurich, Switzerland
Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.	Apr 2014 – July 2015
- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march. - Men of the year 2014 by Time magazine	

Awards

Nobel Prize in Physics	Nov 1921
The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'	
Royal Swedish Academy of Sciences	
www.nobelprize.org/prizes/physics/1921/einstein/biographical	
Max Planck Medal	2029
Awarded for outstanding scientific achievement	
German Physical Society	

Publications

Zur Elektrodynamik bewegter Körper

It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.

Albert Einstein

en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Skills

Physics

Languages

German

Native speaker

English

Fluent

Interests

Physics

Certificates

Machine Learning

Jan 2018

Quantum Computing

Jan 2018

Quantum Information

Jan 2018

Projects

Quantum Computing

Jan 2018 – Jan 2018

Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith